



Federal Board SSC-II Examination

Model Question Paper Computer Science

(Curriculum 2022-23) Scheme of Studies 2009

Section - A (Marks 13)

Time Allowed: 20 minutes

Section-A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent.

Deleting/overwriting is not allowed. Do not use lead pencil.

ROLL NUMBER					
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

Version No.			
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Candidate Sign. _____

Invigilator Sign. _____

Q1. Fill the relevant bubble against each question according to the curriculum. Each part carries one mark.

Sr no.	Question	A	B	C	D	A	B	C	D
i.	Which of the following is an example of system software?	Microsoft Word	Google Chrome	Windows Operating System	Adobe Photoshop	☐	☐	☐	☐
ii.	Why does an operating system use process scheduling?	To execute multiple processes efficiently	To slow down program execution	To prevent applications from running	To increase storage space	☐	☐	☐	☐
iii.	What will be the equivalent of $(25)_{10}$ to $(?)_2$?	11001	10101	10011	11100	☐	☐	☐	☐
iv.	Which of the following logical reasoning type requires solving patterns and shapes?	Boolean Reasoning	Verbal Logical Reasoning	Nonverbal Logical Reasoning	Analytical Reasoning	☐	☐	☐	☐
v.	If a function calculates the average of three numbers but always returns zero, what type of error is likely causing this issue?	Syntax error	Logical error	Compilation error	Run-time error	☐	☐	☐	☐
vi.	Which machine learning method is used for training without labeled data?	Supervised Learning	Unsupervised Learning	Reinforcement Learning	Deep Learning	☐	☐	☐	☐
vii.	Which of the following best explains the purpose of a histogram?	It displays the frequency distribution of numerical data	It compares categorical data	It shows relationships between two variables	It visualizes a database table	☐	☐	☐	☐
viii.	Which cloud computing service model provides virtualized computing resources over the internet?	SaaS (Software as a Service)	PaaS (Platform as a Service)	IaaS (Infrastructure as a Service)	BaaS (Backend as a Service)	☐	☐	☐	☐
ix.	Which technology is used in both robotics and AI personal assistants?	Blockchain	Natural Language Processing (NLP)	3D Printing	Virtual Reality (VR)	☐	☐	☐	☐
x.	How can students help bridge the digital divide?	Promoting digital literacy	Using technology only personally	Ignoring the issue	Restricting internet access	☐	☐	☐	☐
xi.	What is the primary purpose of a data backup?	To speed up the internet	To recover lost data	To increase computer storage	To prevent hacking	☐	☐	☐	☐

xii.	How can graphics effectively convey event details?	Clear fonts, relevant images, key details	Tiny, unreadable text	Overloaded with text and images	abstract imagery without context	○ ○ ○ ○
xiii.	How does bias affect the validity of research findings?	Improves the accuracy of data	Enhances the credibility of the study	Leads to misleading results	Does not affect the validity of findings	○ ○ ○ ○



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Computer Science Model Question Paper

(Curriculum 2022-23) Scheme of Studies 2009

Time allowed: 2.40 hours

Total Marks: 42

Note: Answer all parts from Section 'B' and all questions from Section 'C' on the E-sheet.
Write your answers in the allotted/given spaces.

SECTION – B (Marks 22)

Q.2	Question	Marks	Question		Marks
i.	Define deadlock with an example.	1+1	OR	What is the role of data visualization in the data science process?	2
ii.	A user notices slow performance when running multiple applications on their computer. What system software functions could be responsible for this issue, and what steps can be taken to optimize performance?	1+1	OR	A company wants to develop a system that quickly retrieves customer data from a large database. Which search algorithm would be most suitable for this task, and how does it improve efficiency?	2
iii.	Calculate the Mean Squared Error (MSE) for a given set of predicted and actual values. Actual values (Y_actual): [3, 5, 2, 8, 7] Predicted values (Y_pred): [2.5, 5.3, 1.8, 7.5, 6.8]	2	OR	Convert (2F08) ₁₆ to Octal and Binary.	1+1
iv.	Enlist the scenario in which all the functionality is dealt by the frontend only.	2	OR	Differentiate between System and Application software with examples.	2
v.	Write down any four applications of Real-Time operating system.	2	OR	How does blockchain technology ensure security?	2
vi.	How does fraud detection benefit from machine learning?	2	OR	What are two ways students can responsibly use social media when sharing information?	2
vii.	List any TWO characteristics of an algorithm.	1+1	OR	Write down any TWO advantages of using an online computing platform over an offline one.	2
viii.	Why is it beneficial to break a program into smaller modules? (Give any TWO reasons)	2	OR	How does choosing the right research method impact decision-making in research? Provide an example.	1+1
ix.	What factors should be considered while selecting a machine learning model for data analysis? (Suggest any TWO).	2	OR	How does network monitoring help prevent cyber threats? Give any TWO justifications.	2
x.	How do differences in shape, center, and spread help in understanding the characteristics of a dataset?	2	OR	Why is email filtering important in protecting users? Provide any TWO reasons.	2
xi.	How can AI be used responsibly in law enforcement? Enlist two suggestions.	2	OR	What is phishing, and how does it affect users?	1+1

SECTION – C (Marks 20)

Note: Attempt all questions. Marks for each question are given.

(4 x 05=20)

Q. No.	Question	Marks		Question	Marks
Q.3	Write down the significance of an operating system. Also, describe the following functions: a. File management b. Memory management	1+4	OR	Discuss any five components of digital literacy with one example of each.	5
Q.4	Identify the use of a searching algorithm. Compare the linear search with the binary search algorithm.	1+4	OR	Suggest any five effective strategies to combat online hate speech.	5
Q.5	How do IoT and Blockchain work together to improve cybersecurity? Support your answer with any five suggestions.	5	OR	How are HTML, CSS, JavaScript, Python, and PHP categorized into frontend and backend development?	5

[illegible]